

AVNITH VIJAYRAM

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Education

Carnegie Mellon University

B.S. Computational Biology (School of Computer Science); additional major in Artificial Intelligence

May 2029

GPA: 4.0/4.0

Honors & Awards

- **AHA Scientific Sessions 2026:** poster presentation (first author) on modeling cardiotoxicity with game theory Nov 2026
- **Putnam Mathematical Competition:** ranked 64th out of 4300+; **top scorer at CMU** (scored 60/120) Dec 2025
- **ICPC Latin America Championship:** represented CMU; finished in 9th place Mar 2026
- **Regeneron Science Talent Search:** named a Top 300 Scholar among 2,500 global candidates Jan 2025
- **USA Computing Olympiad (USACO):** achieved Platinum division, the highest tier Dec 2023

Research

Research in Cardiovascular Medicine, Dr. Jennifer Kwan, Yale School of Medicine

Mar 2024 - Present

Identifying Biomarkers of Cardiotoxicity Risk for Cancer Patients

- Built a pipeline in Python and R (BioNERO, iGraph) to construct knowledge graph networks by applying gene coexpression inference and K-means clustering on a large cardiac transcriptomic dataset with 11,386 genes and 22 kinase inhibitors (KI) as cancer treatments
- Identified novel causes behind cardiotoxic side effects of KI treatment by designing a game-theoretic model with a custom Nash-equilibrium payoff score to predict gene expression strategies; accepted as a poster presentation at American Heart Association (AHA) Scientific Sessions 2026 as first author
- Identified disease-linked biological pathways in myocarditis-affected cardiac cells by applying Multi-Omics Factor Analysis (MOFA+) to scRNA-seq and CyTOF data to extract latent factors and genes; co-authored manuscript currently under submission

Undergraduate Researcher, Koes Lab, University of Pittsburgh

Aug 2026 - Present

Diffusion and Flow Matching models for Drug Discovery

- Extending Boltzmann sampling with BoltzNCE to larger systems of molecules using scalable inference-time annealing

Projects

Robot Jousting, HackCMU 2026 Grand Prize

Sep 2026

github.com/avnithv/robot-jousting

- Won Grand Prize at HackCMU 2026 (180+ teams) with a jousting game where players control robotic arms that fight in real time
- Built the software for controlling robotic arms (Python/LeRobot) with sim-to-real transfer of arm trajectories designed in MuJoCo using keyframes and inverse kinematics

GlioGrade: ML diagnostic tool for gliomas

May 2024 - Jun 2025

gliograde.org | [project overview](#), [demo](#), and [documentation](#)

- Achieved 93% accuracy at diagnosing gliomas from 3D MRI images (700+ patients) by training 3D Convolutional Neural Network models in PyTorch; built the tool to run locally to protect patient privacy; used by neuropathologists at 3 clinics; advised by Dr. Kimmo Hatanpää at UT Southwestern
- Built the full pipeline: preprocessing 3D scans (skull-stripping, artifact and rotation correction), mitigating class imbalance through sampling and weighting, and packaging the model into a local web app for clinicians

Experience

CMIMC Math Team, Head

Sep 2025 - Present

- Lead problemsetting for the CMIMC and CMWMC math competitions and manage the problem-writing committee

USA Computing Olympiad, Coach

Oct 2025 - Present

- Proposed, selected, and prepared problems for monthly USACO contests with 5000+ participants

Technical Skills

Languages: Python, C++, Java, R, LaTeX

ML / AI: PyTorch, TensorFlow, NumPy, SciPy, Convolutional Neural Networks (CNNs), Graph Neural Networks, GANs, diffusion models, flow matching

Tools: Git, Linux, Jupyter, Tableau, Power BI, MuJoCo, LeRobot

Domain: scRNA-seq, multi-omics (MOFA+), gene coexpression networks, inverse kinematics, algorithms, data structures, competitive programming